

D1 - Foundation Modules:

1. Planning Logic – Pseudocode & Flowcharts

♦ What is Planning Logic?

Before writing actual code, we **plan the logic** — the sequence of steps a program should perform to achieve a result.

This avoids errors and helps in clear thinking.

♦ Pseudocode

Pseudocode is **plain English-like code** that represents the logic before actual programming.

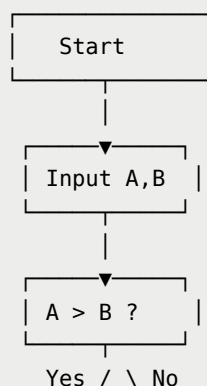
It's not syntax-bound — the goal is clarity.

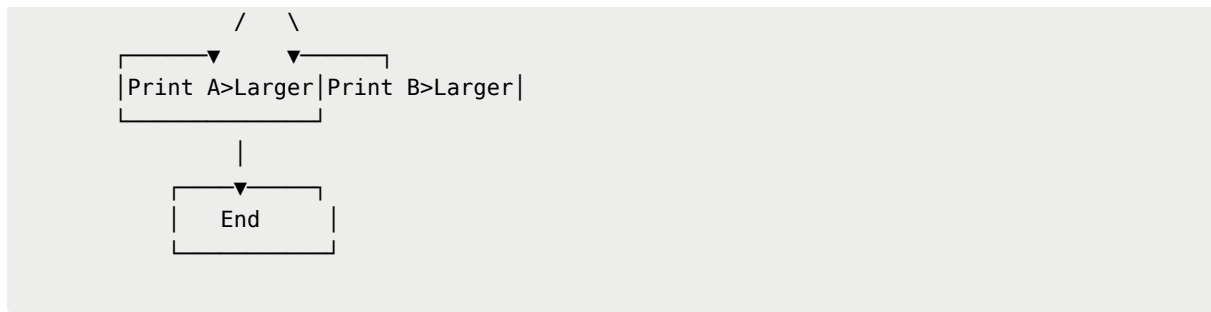
Example: Find the largest of two numbers

```
START
READ A, B
IF A > B THEN
    PRINT "A is larger"
ELSE
    PRINT "B is larger"
END IF
STOP
```

♦ Flowchart

A **flowchart** is a **visual representation** of the logic.





Industry Use:

Used in **software design**, **system architecture**, and **algorithm development** to communicate logic across teams before implementation.

2. Programming Basics – What is Programming? Types of Languages

♦ What is Programming?

Programming is **instructing a computer** to perform specific tasks by writing **logical, structured commands** in a programming language.

♦ Types of Programming Languages

Type	Description	Example
Low-Level	Closer to hardware; difficult but fast	Assembly
High-Level	Human-readable; portable	Python, Java, C++
Procedural	Step-by-step instructions	C
Object-Oriented (OOP)	Uses objects & classes	Java, Python
Scripting	Automates small tasks	Python, JavaScript
Functional	Based on functions & recursion	Haskell, Scala

Industry Use:

Python is preferred for **data**, **AI**, **automation**, Java for **enterprise apps**, and JavaScript for **web development**.

3. Syntax & Structure – Variables, Data Types, Operators

♦ Syntax

Syntax = **rules** of the programming language.

Example (Python):

```
print("Hello, World!")    # Correct
Print("Hello, World!")    # Error (case-sensitive)
```

◆ Variables

A **variable** stores a value in memory.

```
name = "Dinesh"
age = 25
```

◆ Data Types

Type	Example	Description
int	10	Integer number
float	10.5	Decimal number
str	"Hello"	Text
bool	True/False	Logical
list	[1,2,3]	Collection

◆ Operators

Type	Example	Description
Arithmetic	+, -, *, /	Math ops
Comparison	>, <, ==, !=	Compare values
Logical	and, or, not	Combine conditions
Assignment	=, +=, -=	Assign values

Example:

```
a, b = 10, 5
print(a + b)    # 15
print(a > b)    # True
```

4. Control Structures – if, else, loops (for, while)

These control how code executes — **decision-making & repetition**.

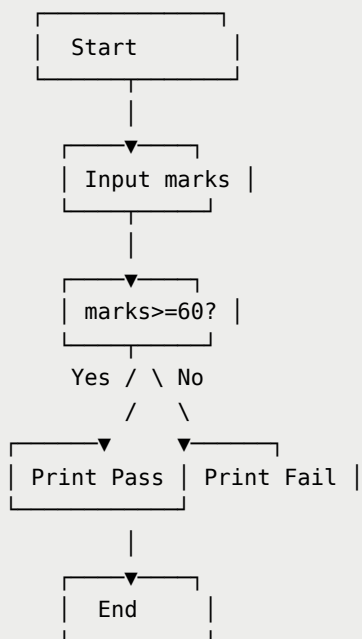
♦ if-else (Decision)

```
marks = 75
if marks >= 60:
    print("Pass")
else:
    print("Fail")
```

Pseudocode:

```
IF marks >= 60 THEN
    PRINT "Pass"
ELSE
    PRINT "Fail"
```

Flowchart:



♦ Loops (Repetition)

For Loop (fixed repetition)

```
for i in range(1, 6):  
    print("Iteration:", i)
```

→ Prints 1 to 5

While Loop (condition-based)

```
count = 1  
while count <= 5:  
    print("Count:", count)  
    count += 1
```

Industry Use:

Used for **data iteration**, **user input validation**, **repetitive tasks**, etc.

5. Functions & Modular Programming

♦ Why Functions?

Functions **modularize** code — make it reusable, organized, and easy to test.

Definition:

```
def greet(name):  
    return "Hello, " + name
```

Calling:

```
message = greet("Karthik")  
print(message)
```

Output:

```
Hello, Karthik
```

♦ Parameters and Return Values

- **Parameters:** inputs to the function
- **Return value:** output from the function

```
def add(a, b):  
    return a + b  
  
result = add(10, 20)  
print(result)  # 30
```

Pseudocode:

```
FUNCTION add(a, b)  
    RETURN a + b  
END FUNCTION
```

♦ Modular Programming

Breaking the program into **smaller, independent modules** (functions or files).

Example Project: Calculate Student Grades

```
def input_marks():  
    return [85, 78, 92]  
  
def calculate_average(marks):  
    return sum(marks) / len(marks)  
  
def display_result(avg):  
    if avg >= 60:  
        print("Pass")  
    else:  
        print("Fail")  
  
# Main module  
marks = input_marks()  
average = calculate_average(marks)  
display_result(average)
```

 Reusable

- ✓ Easy to debug
 - ✓ Industry-standard structure
-

Summary Table

Concept	Purpose	Example
Pseudocode	Plan logic	Steps to find largest number
Flowchart	Visual logic	Diamonds = decision
Programming Basics	Instruct computer	Python, Java, etc.
Variables & Data Types	Store data	name = "John", age = 25
Operators	Perform actions	+, -, ==
Control Structures	Decision & loops	if, for, while
Functions	Reusable blocks	def add(a, b)
